

Business Data Management Capstone Project

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Project Title -

Maximizing Revenue and Improving Inventory Management in a
Bakery: A Data Driven Approach

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Organization Background

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- Golden Crust is a bakery based in Kanpur, offering a variety of baked and confectionery items.
- Run by Mr. Divyanshu Gupta, who manages daily operations.
- Products include cakes, pastries, dry cakes, snacks, and more.
- Focused on quality and customer satisfaction.



Problem Statement

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- Lack of sales tracking, leading to poor financial visibility.
- Inefficient inventory management, causing stockouts or overstocking.
- Potential revenue loss due to unstructured sales and inventory control.

Data Collection & Methodology

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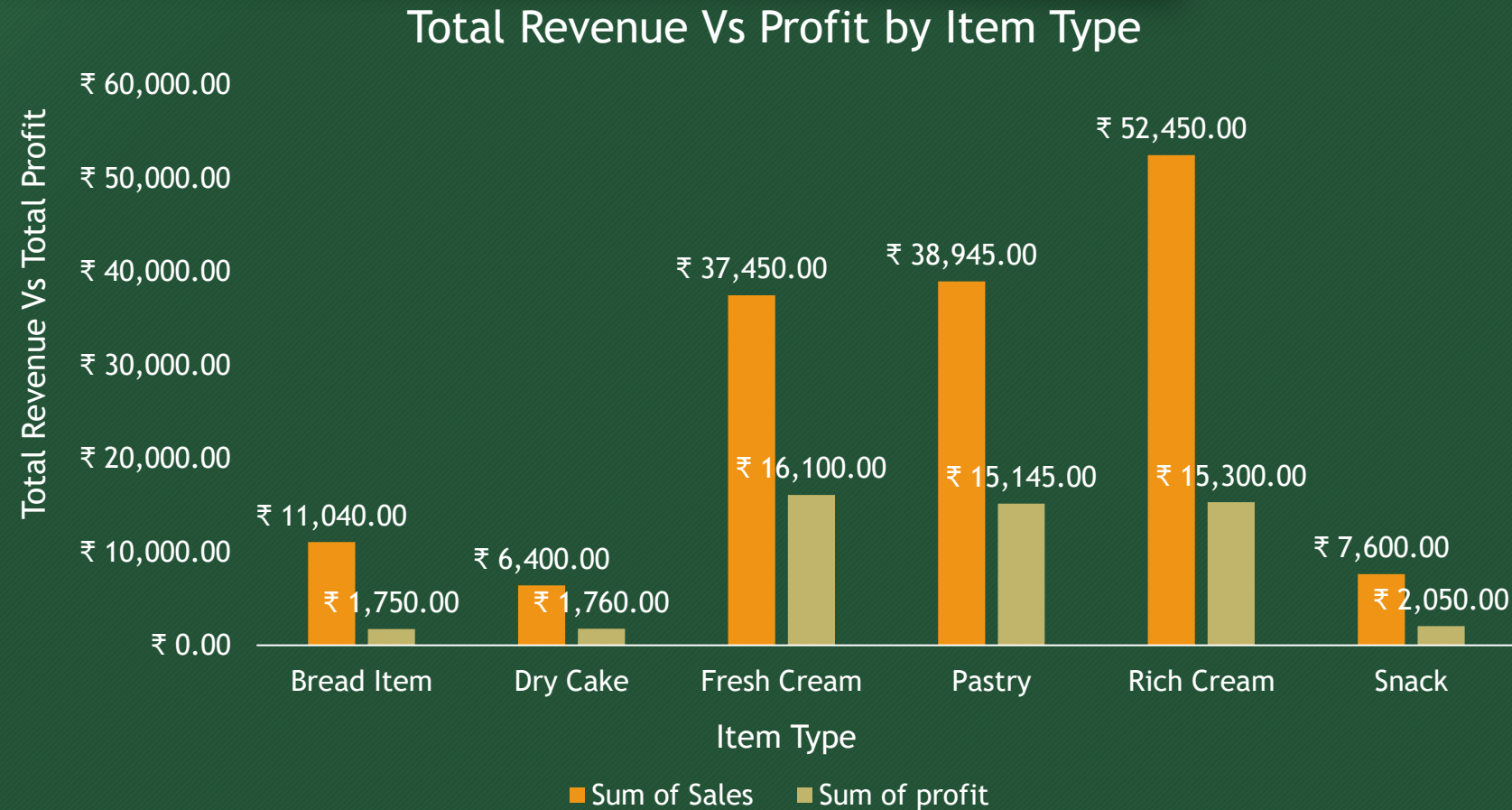
- Collected manually over 30 days at Golden Crust.
- Recorded daily sales, inventory levels, prices, dates, and SKUs.
- Data entered and maintained using **Microsoft Excel**.
- Ensured consistency through close coordination with the owner.

Data Insights

(Total Revenue Vs Profit By Item Type)

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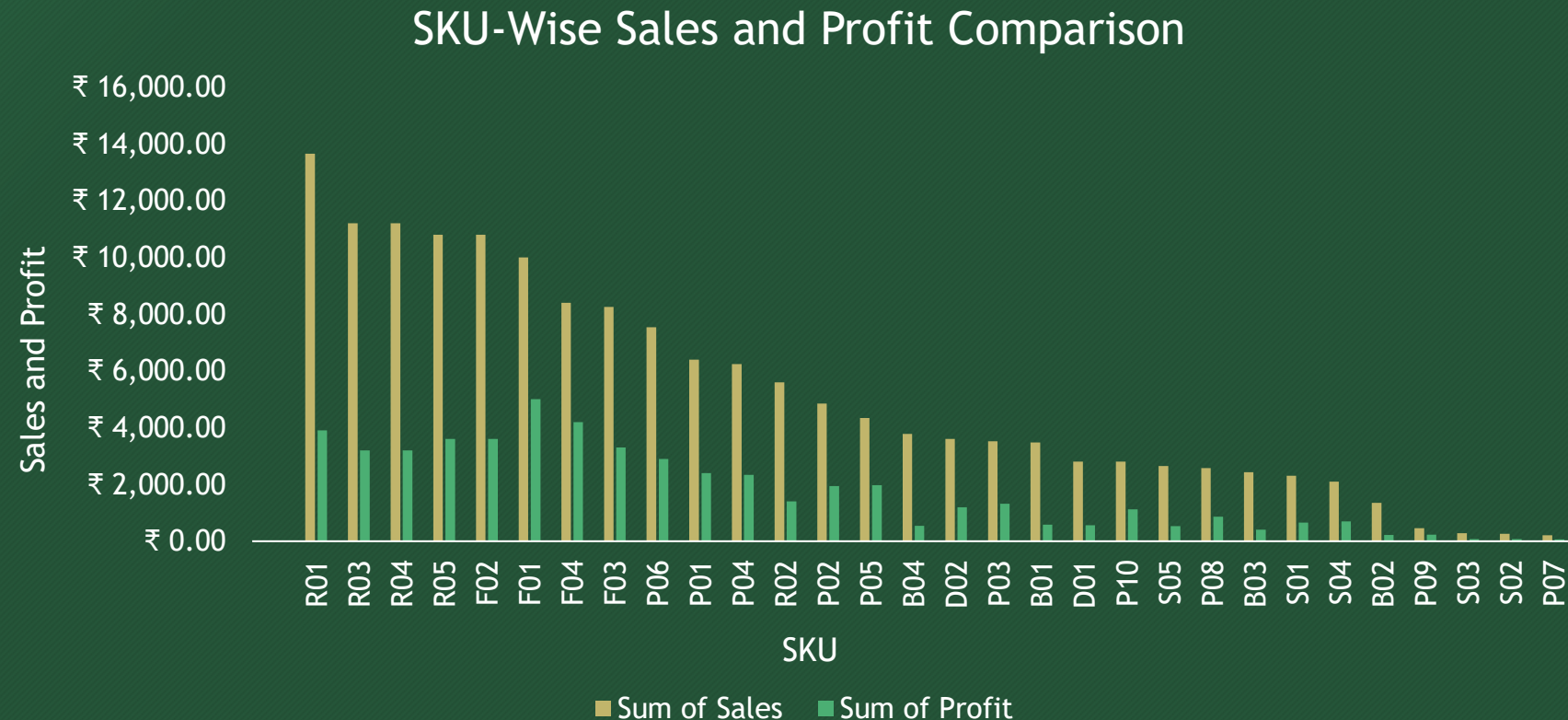
- **Rich Cream** items generated the highest revenue (₹52,450), followed by **Pastries** and **Fresh Cream** which generated higher profits.
- **Dry Cake** and **Bread Items** had the **lowest profit margins**, despite moderate sales.
- **Snack** items, though low in revenue, showed **decent profit per sale**, suggesting potential for targeted promotion.



Data Insights (SKU-Wise Sales and Profit Comparison) (ABC Analysis)

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- **Class A (Top 70% of Sales):**
Critical products that must always be in stock (e.g., R01, R03, R04, R05, F02, F01, F04, F03).
- **Class B (Next 20% of Sales):**
Moderate stock levels.(e.g., P02, P05, B04, D02).
- **Class C (Final 10% of Sales):**
Low-priority items which should be minimized or discontinued (e.g., S05, P08, B03, S01).



Data Insights (Weekly Sales Trend)

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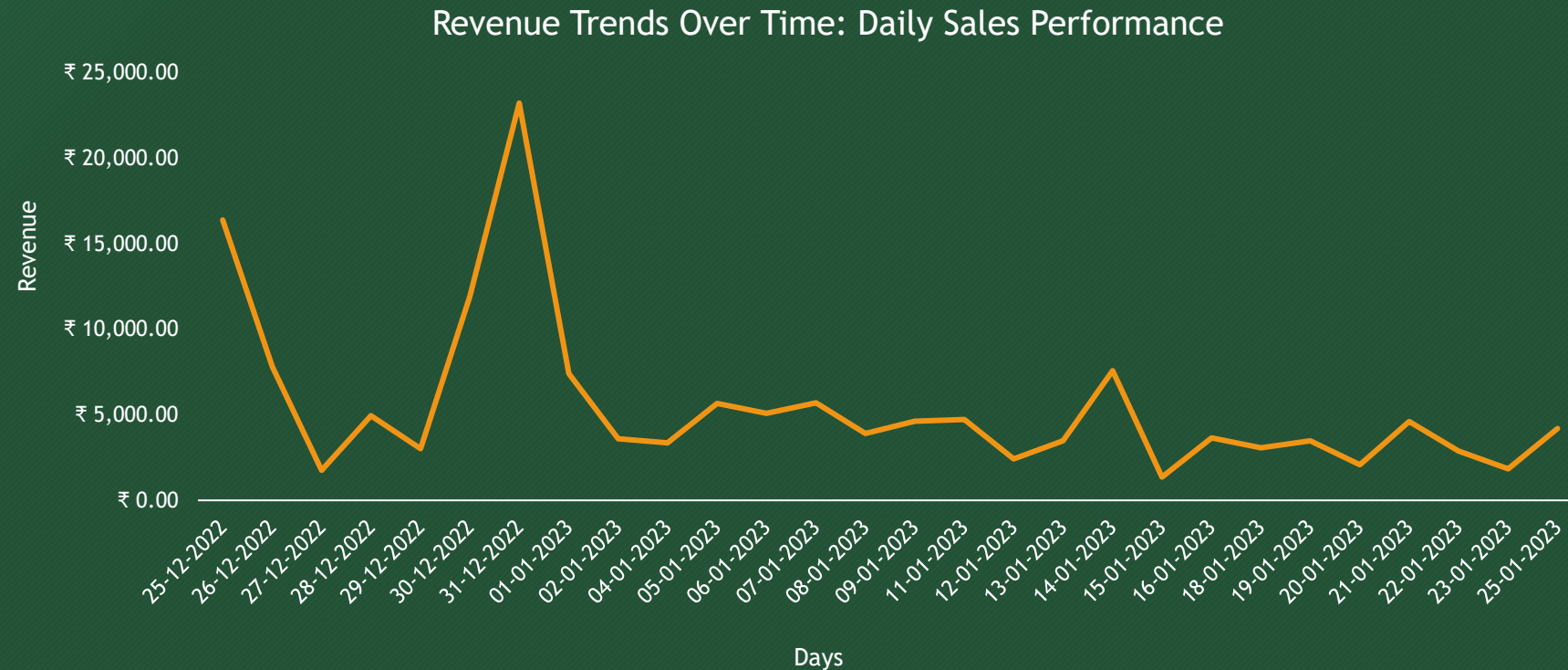
- Monday and Wednesday show moderate sales (₹21,550 and ₹20,295, respectively).
- Lowest Sales on Tuesday (Off Day) suggest minimal customer engagement midweek.
- Thursday (₹14,600) has relatively lower sales compared to the previous days.
- Friday (₹22,590) sees an upward trend leading into the weekend.
- Peak Sales on Saturday (₹41,145) and Sunday (₹31,955) which indicates high weekend demand.



Data Insights (Daily Sales Performance)

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- **Year-End Surge (Dec 30-31):** Major spike in revenue during New Year festivities, reflecting high seasonal demand.
- **Post-Holiday Dip (From Jan 1):** Sales drop sharply after New Year and stabilize, showing a return to normal daily demand.
- **Mid-January Variations (Around Jan 7 & 20):** Small revenue peaks possibly linked to weekends.



Strategic Recommendations

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1. Optimize Pricing & Costs

- Renegotiate supplier rates for low-margin, high-volume SKUs (e.g. R01)
- Introduce premium variants on high demand products. (e.g. Rich Cream, Pastries)

2. ABC-Driven Inventory Control

- Maintain 7-day buffer stock for Class A products; minimize Class C holdings
- Adopt just-in-time ordering for perishables to reduce waste. (e.g. Rich cream 3-4kg)

3. Targeted Promotions & Dynamic Pricing

- Weekend Specials (e.g. “Saturday Mega Bundles”) and midweek loyalty discounts
- Offer exclusive product variants during festive periods to capitalize on increased footfall.

4. Operational & Retention Tactics

- Increase weekend staffing; use Tuesdays for audits and training
- Launch a loyalty program (e.g. “Buy 5 Rich Creams, Get 1 Free”) and QR-code feedback

Implementation & Impact

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1. Premium Rich Cream Variants on Weekends:

The bakery began offering premium versions of Rich Cream cakes during weekends. This attracted more customers and led to an increase in overall sales.

2. Training & Auditing on Alternate Tuesdays:

Every second Tuesday is now reserved for staff training and stock audits, helping improve efficiency and reducing errors in operations.

3. Discontinuation of Low-Performing Items (P09 & P07):

Items from Class C, such as P09 and P07, were discontinued to reduce overstocking and minimize product spoilage. The remaining Class C items are now stocked in limited quantities, leading to better inventory control and reduced waste.

Conclusion & Thank You

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- This project delivered clear, data-driven strategies to boost Golden Crust's revenue and streamline inventory management.
- Precision in data collection and cleaning was critical—small mistakes can lead to big missteps in analysing and decision-making.
- ABC classification helped categorize products based on their sales and importance. This turned raw data into clear insights, allowing us to focus on high-performing items, reduce low-priority stock, and make smarter inventory and sales decisions.